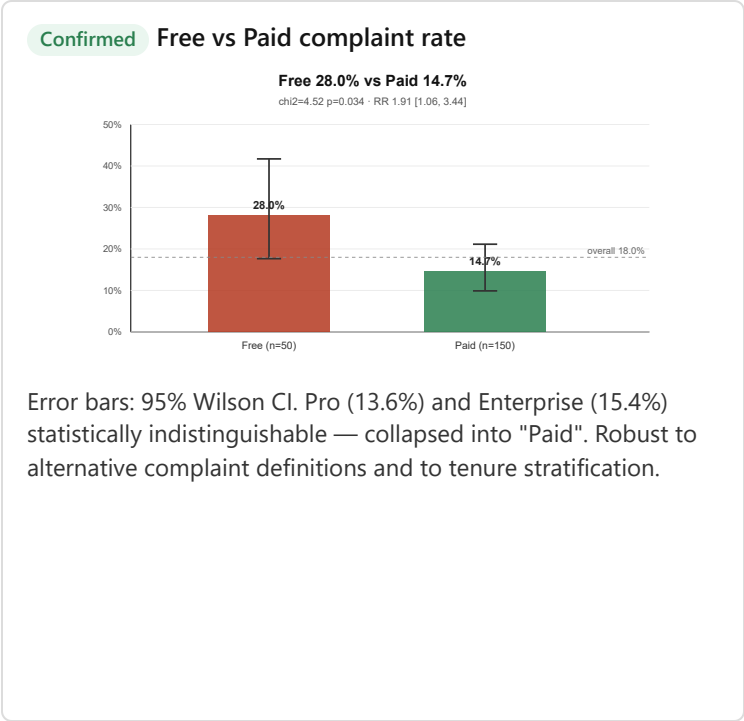


Who files complaints — and can we predict it?

Analyst: liati · Snapshot: 2024-06-17 → 2025-06-17 · N = 200 users, 36 complainers (18.0% baseline)

Answer: Only one observable attribute reliably distinguishes complainer users in this dataset — **plan = 'free'**. Free users complain at **~1.9× the rate of paid users** (28.0% vs 14.7%, χ^2 p = 0.034, RR 1.91 [95% CI 1.06, 3.44]). Role, tenure, and industry do not distinguish. Pre-complaint engagement *cannot be tested* because event-data coverage is only ~90 days of the 365-day window.

Supporting findings



Refuted Role & tenure don't distinguish

Attribute	Max lift	p-value
Role (admin/analyst/viewer)	1.16×	0.74
Tenure (4 buckets)	1.32× (<90d)	0.30
Industry (7 buckets)	–	underpowered

No segment meets the 1.3× lift + p < 0.05 bar. Don't use these for targeting.

Inconclusive Engagement not testable

The `events` table covers only 2025-03-07 → 2025-06-06 (~90 days). 18 of 36 complainers had their first complaint before events started — forcing structural zeros. In the n=14 subset with valid coverage, complainers have *higher* engagement (median 5 vs 2, z=+2.27) — too small for a decision-grade claim.

Recommendations

- **Deploy** `plan = 'free'` as the primary watch-list rule. Precision 28%, recall 39%, lift 1.56× over baseline. Simple, unconfounded, reproducible.
- **Do not deploy any event-based rule yet.** `ev_14 = 0` has below-baseline precision (0.94×). The higher-lift rules involving `ev_30 = 0` are contaminated by the event-coverage gap — mark as experimental.
- **Extend** `events` coverage to ≥6 months overlapping the complaint window. Single highest-leverage data improvement. Would unlock H2 (engagement) testing.
- **Do not use role, tenure, or industry as watch-list attributes.** None separate complainers from non-complainers.
- **Re-run in 6 months** to confirm whether the Free-plan effect is structural or a 2025-specific period effect.

Caveats & limitations

- **The "12-month" window is effectively 6 months.** 34 of 36 complaints occurred in 2024-12-17 → 2025-06-17. Findings describe that period; generalisation beyond it is not supported.
- **Event-data coverage gap.** 90 days of events inside a 365-day analysis window. Added to `.cursor/learning/known_issues.md` — future analyses should check `MIN/MAX(occurred_at)` during Phase 2 QA.
- **Small n.** 36 complainers is at the edge of what supports segmented claims. Only the plan comparison clears the $1.3\times$ lift + $p<0.05$ bar with $n\geq 30$.
- **Methodological correction during validation.** An earlier draft reported a "disengagement" finding that was driven by (a) a SQL rank-sum tie-handling bug and (b) the event-coverage gap. The corrected analysis withdraws that claim; full audit trail in `results/validation/validation.md`. Two new entries in `known_issues.md`.
- **Industry analysis underpowered** — ~7 accounts per industry bucket is too few to detect industry-level differences.

What we couldn't answer

- **Is pre-complaint engagement a signal?** Not testable at current event coverage.
- **Is the Free-plan effect structural or period-specific?** Single 6-month observation; needs a second snapshot or a longer time series.
- **Why does the paid stratum show a tenure effect but the free stratum doesn't?** Observed as a side finding (paid 17%→11% newer-to-older; free 28%/28%); scoped out for follow-up.
- **Why does complaint activity concentrate in late 2024–2025?** Could be real (product change, base growth) or reporting (support-ticket logging change). Cannot distinguish from this data alone.

Evidence scorecard: 1 Moderate finding (Free plan, narrowed to 6-month window), 2 Refuted (role, tenure), 1 Inconclusive (engagement), 1 Withdrawn (earlier disengagement reframe). See full `report.html` in this folder for charts, query index, and the complete validation trail.